



Finite-sample guarantees for logistic regression with Gaussian design

Minimax Optimal Rate, Maximum Likelihood Estimator, and Gradient Descent

Junren Chen

Department of Statistics, Columbia University

CUNY GC Harmonic Analysis and PDE

September 18, 2026

Joint work with Arya Mazumdar (UC San Diego)

Model and estimators

Gaussian-design logistic model

$$x_i \stackrel{\text{iid}}{\sim} N(0, I_d), \quad y_i \mid x_i \sim \text{Bernoulli}(s(x_i^\top \theta^*)), \quad i = 1, \dots, n$$

$$s(z) = \frac{1}{1 + e^{-z}}, \quad \theta^* = Rv^*, \quad \|v^*\| = 1, \quad R \geq 1.$$

- R is the signal strength / inverse temperature.
- We track high-probability errors with explicit dependence on (n, d, R) .

Logistic loss: MLE and GD

$$L_n(\theta) = \frac{1}{n} \sum_{i=1}^n \left\{ \log(1 + e^{x_i^\top \theta}) - y_i x_i^\top \theta \right\}.$$

$$\hat{\theta}_{\text{MLE}} \in \arg \min_{\theta} L_n(\theta),$$

$$\theta_{t+1} = \theta_t - \eta \nabla L_n(\theta_t)$$

Minimax Optimal Rate

$$\mathfrak{R}_n(R_0) = \inf_{\hat{\theta}} \sup_{\theta^* \in \Theta_{R_0}} \mathbb{E}_{\theta^*} \left\| \hat{\theta} - \theta^* \right\|_2.$$

where

$$\Theta_{R_0} = \{\theta : R_0 \leq \|\theta\| \leq 2R_0\}.$$

Asymptotic Viewpoints

Let $L(\theta) = \mathbb{E}L_n(\theta)$

Classical asymptotics

Fixed d and fixed R , with $n \rightarrow \infty$:

$$\begin{aligned}\hat{\theta}_{\text{MLE}} \text{ exists w.p. } &\rightarrow 1, \\ \|\hat{\theta}_{\text{MLE}} - \theta^*\|_2 &= O_{\mathbb{P}}(n^{-1/2}).\end{aligned}$$

More sharply,

$$2n\{L(\hat{\theta}_{\text{MLE}}) - L(\theta^*)\} \Rightarrow \chi^2(d).$$

Limitation: it does not say how large finite n must be relative to (d, R) .

High-dimensional asymptotics

Fixed signal $R = b$, with $d, n \rightarrow \infty$ and $d/n \rightarrow \gamma$:

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{P}(\text{MLE exists}) \\ = \begin{cases} 1, & \gamma < h(b), \\ 0, & \gamma > h(b). \end{cases}\end{aligned}$$

This captures the interaction between dimension and signal, but still keeps R fixed and is asymptotic.

These results are precise but do not apply to finite samples, do not allow growing d or R

Existence and excess risk of the MLE in Finite Samples

General finite-sample theory

Chinot–Lecué–Lerasle. If $n \gtrsim e^{cR}d$, then the MLE exists and

$$L(\hat{\theta}_{\text{MLE}}) - L(\theta^*) \lesssim \frac{e^{cR}d}{n}.$$

Robust/general, but exponential in R .

Gaussian-design refinement

Ostrovskii–Bach. If $n \gtrsim R^8d$ up to logarithms, then

$$L(\hat{\theta}_{\text{MLE}}) - L(\theta^*) \lesssim \frac{R^3d}{n}$$

up to a deviation factor.

More recent benchmark. (Chardon–Lerasle–Mourtada, 2024) Under the well-specified Gaussian logit model, if $n \gtrsim Rd$, then MLE exists and satisfies

$$L(\hat{\theta}_{\text{MLE}}) - L(\theta^*) \lesssim \frac{d}{n};$$

If $n \lesssim Rd$, then

$$\mathbb{P}(\hat{\theta}_{\text{MLE}} \text{ does not exist}) \rightarrow 1$$

Finite-sample estimation performance of the MLE

(Kuchelmeister–van de Geer, 2024)
(probit)

Under $n \gtrsim Rd \log n$,

$$\begin{aligned}\|\hat{v}_{\text{MLE}} - v^*\| &\lesssim \sqrt{\frac{d \log n}{nR}}, \\ |\hat{R}_{\text{MLE}} - R| &\lesssim \sqrt{\frac{R^3 d \log n}{n}}.\end{aligned}$$

(Chardon–Lerasle–Mourtada, 2024)
(logit)

Under $n \gtrsim Rd$,

$$\begin{aligned}\|\hat{v}_{\text{MLE}} - v^*\| &\lesssim \sqrt{\frac{d}{nR}}, \\ |\hat{R}_{\text{MLE}} - R| &\lesssim \sqrt{\frac{R^3 d}{n}}.\end{aligned}$$

Previous best-known MLE estimation error rate:

$$\left\| \hat{\theta}_{\text{MLE}} - \theta^* \right\| \lesssim \sqrt{\frac{R^3 d}{n}}$$

Direction improves with R , while the norm bound worsens with R and carries $\sqrt{d}$.

Intuition: $R \rightarrow \infty \implies$ halfspace learning, where norm estimation is not possible

Open Questions in Parameter Estimation

Known MLE upper bound

Write $\hat{\theta}_{\text{MLE}} = \hat{R}_{\text{MLE}} \hat{v}_{\text{MLE}}$, if $n \gtrsim Rd$,

$$\|\hat{v}_{\text{MLE}} - v^*\| \lesssim \sqrt{\frac{d}{nR}},$$

$$|\hat{R}_{\text{MLE}} - R| \lesssim \sqrt{\frac{R^3 d}{n}},$$

$$\|\hat{\theta}_{\text{MLE}} - \theta^*\|_2 \lesssim \sqrt{\frac{R^3 d}{n}}$$

Known Lower Bound (Hsu–Mazumdar, 2024)

Under $n \gtrsim Rd$ (MLE-existence regime),

$$\inf_{\hat{v}} \sup_{v^* \in S^{d-1}} \mathbb{E}_{\theta^*} \|\hat{v} - v^*\|_2 \gtrsim \sqrt{\frac{d}{nR}},$$

$$\inf_{\hat{\theta}} \sup_{\theta^* \in RS^{d-1}} \mathbb{E}_{\theta^*} \|\hat{\theta} - \theta^*\|_2 \gtrsim \sqrt{\frac{Rd}{n}}.$$

Open questions & our contributions (Chen–Mazumdar, 2026b)

- Are the MLE estimation error rates tight? **No, improved to** $\tilde{O}(\sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3}{n}} + \frac{R^2 d}{n})$
- What is the minimax optimal error rate? $\Theta(\sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3}{n}})$ (**non-MLE estimator**)

Gradient descent: prior results and open question

$$\theta_{t+1} = \theta_t - \frac{\eta}{n} \sum_{i=1}^n (s(x_i^\top \theta_t) - y_i) x_i, \quad t = 0, 1, 2, \dots$$

Prior results

- **Classical convex optimization:** (Nesterov, 2018) if

$$\eta \leq \frac{2}{\beta}, \quad \beta = \sup_{v \in \mathbb{R}^d} \|\nabla^2 L_n(v)\|_{op},$$

then $L_n(\theta_t) - L_{n,\min} = O(T^{-1})$

- **Estimation guarantees:** prior results do not give an explicit finite-sample ℓ_2 error bound to θ^* under Gaussian design (Kakade et al., 2011)
- **Large step sizes:** recent works show acceleration via large-stepsize GD under **separable** data (Wu–Bartlett–Telgarsky–Yu, 2024)

Open questions & our contributions (Chen–Mazumdar, 2026a)

- Finite-sample estimation performance?

$$\tilde{O}\left(\sqrt{\frac{R^5 d}{n}}\right) \text{ under } n = \tilde{\Omega}(R^6 d)$$

- Any faster convergence?

Global linear convergence in parameter

- Any acceleration under large stepsize?

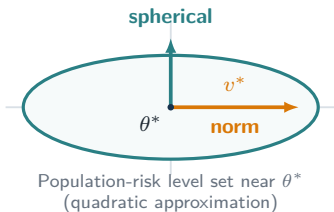
a local linear convergence with sharper contraction factor

Preliminaries

For $r \geq 0$ and $g \sim N(0, 1)$, define

$$q(r) := \mathbb{E}[g s(rg)], \quad q'(r) = \mathbb{E}[g^2 s'(rg)] > 0.$$

Thus q is strictly increasing and invertible on its range; $q(r) \asymp 1$ for $r \geq 1$.



Gaussian identity ($\|v\|_2 = 1$):

$$\mathbb{E}[s(rx^\top v)x] = q(r)v.$$

Population Hessian at θ^* :

$$\lambda_{\min} = q'(R) \asymp R^{-3},$$

$$\lambda_{\max} = \frac{q(R)}{R} \asymp R^{-1}.$$

Roadmap and contributions

PART I

Minimax lower bound

$$\sqrt{Rd/n} + \sqrt{R^3/n}$$

PART II

Improved MLE upper bound

$$\sqrt{Rd/n} + \sqrt{R^3/n} + R^2d/n$$

PART III

Minimax upper bound

$$\sqrt{Rd/n} + \sqrt{R^3/n}$$

PART IV

GD

linear convergence to $\tilde{O}(\sqrt{R^5d/n})$

- Parts I+III: minimax optimal rate is $\sqrt{Rd/n} + \sqrt{R^3/n}$
- Part II: Best-known MLE error rate $\sqrt{Rd/n} + \sqrt{R^3/n} + R^2d/n$ (conjectured to be tight)
- Part IV: Fast, global linear convergence to statistical error rate

Minimax lower bound: direction and norm

Annulus $\Theta_{R_0} = \{\theta \in \mathbb{R}^d : R_0 \leq \|\theta\| \leq 2R_0\}$

If $R_0 \geq 1$ and $n \gtrsim R_0 + d/R_0$, then every estimator obeys

$$\inf_{\hat{\theta}} \sup_{\theta^* \in \Theta_{R_0}} \mathbb{E}_{\theta^*} \left\| \hat{\theta} - \theta^* \right\|_2 \gtrsim \underbrace{\sqrt{\frac{R_0 d}{n}}}_{\text{direction estimation}} + \underbrace{\sqrt{\frac{R_0^3}{n}}}_{\text{norm estimation}}.$$

Restrict the parameter space to

$$\theta^* = R e_1, \quad R \in [R_0, 2R_0].$$

Even if the direction is revealed,

$$\inf_{\hat{R}} \sup_{R \in [R_0, 2R_0]} \mathbb{E}_{R e_1} \left| \hat{R} - R \right| \gtrsim \sqrt{\frac{R_0^3}{n}}.$$

Difficulty of norm estimation

Le Cam two-point method

Reveal the direction v^* . Compare $\theta_0 = R_0 v^*$, $\theta_1 = (R_0 + h)v^*$.

Le Cam says:

$$\inf_{\widehat{R}} \sup_{j=0,1} \mathbb{E}_{\theta_j} \left| \widehat{R} - R_j \right| \geq \frac{h}{2} \left(1 - \text{TV}(P_0^{\otimes n}, P_1^{\otimes n}) \right).$$

So we need to choose h large enough while keeping $\text{TV}(P_0^{\otimes n}, P_1^{\otimes n}) \leq \frac{1}{2}$

Key calculation

The proof shows

$$D_{\text{KL}}(P_0^{\otimes n} \parallel P_1^{\otimes n}) \lesssim nh^2/R_0^3.$$

Choose

$$h = \kappa \sqrt{R_0^3/n}$$

with small enough κ to ensure

$$D_{\text{KL}}(P_0^{\otimes n} \parallel P_1^{\otimes n}) \leq 1 \xRightarrow{\text{ Pinsker }} \text{TV}(P_0^{\otimes n}, P_1^{\otimes n}) \leq \frac{1}{2}$$

Improved finite-sample MLE rate

Full MLE error bound

When $n \gtrsim Rd$, with high probability, the MLE exists and satisfies

$$\left| \hat{R}_{\text{MLE}} - R \right| \lesssim \sqrt{\frac{R^3 \log(nR)}{n}} + \frac{R^2 d \log(nR)}{n}.$$

$$\left\| \hat{\theta}_{\text{MLE}} - \theta^* \right\|_2 \lesssim \sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3 \log(nR)}{n}} + \frac{R^2 d \log(nR)}{n}.$$

Ignoring logarithmic factors, our norm estimation rate is a strict improvement on $O\left(\sqrt{\frac{R^3 d}{n}}\right)$ due to Chardon et al. (2024):

$$n \geq Rd \implies \sqrt{\frac{R^3}{n}} \leq \sqrt{\frac{R^3 d}{n}}, \quad \frac{R^2 d}{n} \leq \sqrt{\frac{R^3 d}{n}}$$

Looseness in existing analysis

MLE upper bound from Chardon–Lerasle–Mourtada (2024)

When $n \gtrsim Rd$,

$$\|\hat{v}_{\text{MLE}} - v^*\|_2 \lesssim \sqrt{\frac{d}{nR}}, \quad \left| \hat{R}_{\text{MLE}} - R \right| \lesssim \sqrt{\frac{R^3 d}{n}}.$$

Norm estimation

The norm bound comes from a coarse conversion:

$$\begin{aligned} \left| \hat{R}_{\text{MLE}} - R \right| &\leq \left\| \hat{\theta}_{\text{MLE}} - \theta^* \right\|_2 \\ &\leq \underbrace{\sqrt{\lambda_{\max}(H^{-1})}}_{=\sqrt{\lambda_{\min}(H)} \asymp R^{3/2}} \times \underbrace{\left\| \hat{\theta}_{\text{MLE}} - \theta^* \right\|_H}_H \lesssim \sqrt{\frac{R^3 d}{n}}. \\ &:= \sqrt{\underbrace{(\hat{\theta}_{\text{MLE}} - \theta^*)^\top H (\hat{\theta}_{\text{MLE}} - \theta^*)}_{\lesssim \sqrt{d/n}}} \end{aligned}$$

Reduction

First-order optimality $\Rightarrow$ norm error

$$\nabla L_n(\hat{R}_{\text{MLE}}\hat{v}_{\text{MLE}}) = 0 \quad \implies \quad \langle \nabla L_n(\hat{R}_{\text{MLE}}\hat{v}_{\text{MLE}}), \hat{v}_{\text{MLE}} \rangle = 0.$$

$$|\hat{R}_{\text{MLE}} - R| \lesssim R^3 \left\{ \|\hat{v}_{\text{MLE}} - v^*\|_2^2 + |S_n(\hat{R}_{\text{MLE}}, \hat{v}_{\text{MLE}})| \right\}.$$

$$S_n(r, v) := \langle \nabla L_n(rv) - \mathbb{E} \nabla L_n(rv), v \rangle$$

Chardon et al. already gives

$$\|\hat{v}_{\text{MLE}} - v^*\|_2 \lesssim \sqrt{\frac{d}{nR}}.$$

Hence

$$R^3 \|\hat{v}_{\text{MLE}} - v^*\|_2^2 \lesssim \frac{R^2 d}{n}.$$

Localized covering (bounding $S_n(\hat{R}_{\text{MLE}}, \hat{v}_{\text{MLE}})$)

Localize by the existing MLE bound

Let $\hat{v}_{\text{MLE}} \in \mathcal{V} := \{v : \|v - v^*\|_2 \lesssim \sqrt{d/(nR)}\}$, $\hat{R}_{\text{MLE}} \in [R/2, 2R]$,

$$\begin{aligned} R^3 |S_n(\hat{R}_{\text{MLE}}, \hat{v}_{\text{MLE}})| &\leq R^3 \sup_{r \in [R/2, 2R]} \sup_{v \in \mathcal{V}} |S_n(r, v)| \\ &\leq R^3 \sup_{r \in [R/2, 2R]} |S_n(r, v^*)| + R^3 \sup_{r \in [R/2, 2R]} \sup_{v \in \mathcal{V}} |S_n(r, v) - S_n(r, v^*)|. \end{aligned}$$

Bernstein inequality (moment conditions) + covering argument

$$R^3 \sup_r |S_n(r, v^*)| = \tilde{O}\left(\sqrt{\frac{R^3}{n}}\right) \quad (1)$$

$$R^3 \sup_{r, v} |S_n(r, v) - S_n(r, v^*)| = \tilde{O}\left(\frac{R^2 d}{n}\right). \quad (2)$$

Detail for (2):

- Bernstein: $R^3 |S_n(r, v) - S_n(r, v^*)| = \tilde{O}(\frac{R^2 \sqrt{d}}{n})$
- Covering: $R^3 \sup_{r, v} |S_n(r, v) - S_n(r, v^*)| = \tilde{O}(\frac{R^2 d}{n})$.

Is $R^2 d/n$ intrinsic to the MLE?

High-dimensional Asymptotics

- Sur–Candès (2019): “MLE tends to overestimate the effect size.”
(e.g., $\theta^* \leftrightarrow 1.1\theta^*$, $s(x_i^\top \theta^*) \leftrightarrow s(1.1x_i^\top \theta^*)$)
- Zhao–Sur–Candès (2022): in the high-dimensional limit $d/n \rightarrow \gamma$, the MLE norm satisfies

$$\hat{R}_{\text{MLE}} \xrightarrow{\text{a.s.}} r_{\text{MLE}}(\gamma, R).$$

Then take the low-dimensional limit $\gamma \rightarrow 0^+$.

- Our calculation gives

$$\lim_{\gamma \rightarrow 0^+} \frac{r_{\text{MLE}}(\gamma, R) - R}{\gamma} = \frac{1}{2Rq'(R)} \asymp R^2.$$

Hence

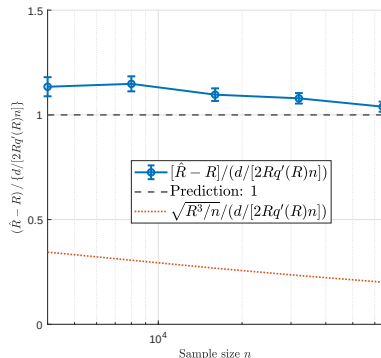
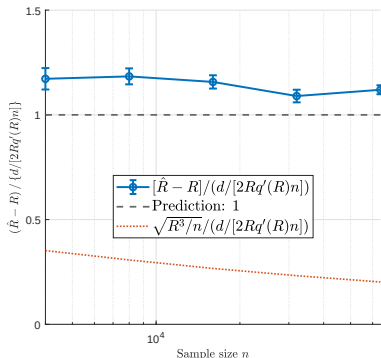
$$r_{\text{MLE}}(\gamma, R) - R \sim \frac{\gamma}{2Rq'(R)} \quad \xrightarrow{d/n \rightarrow \gamma} \quad \hat{R}_{\text{MLE}} - R \sim \frac{d}{2Rq'(R)n} \asymp \frac{R^2 d}{n}.$$

- Conjecture:

$$\hat{R}_{\text{MLE}} - R \gtrsim \frac{R^2 d}{n} \quad \text{under } n \gtrsim Rd$$

Is $R^2 d/n$ intrinsic to the MLE? Numerical Simulations

- Left: $R = 5$, $d = 0.5n^{0.7}$. Then $R^2 d/n \asymp n^{-0.3}$, while $\sqrt{R^3/n} \asymp n^{-0.5}$.
- Right: $R = n^{0.15}$, $d = 0.5n^{0.7}$. Then $R^2 d/n \asymp n^{-0.1}$, while $\sqrt{R^3/n} \asymp n^{-0.275}$.



Is the extra R^2d/n term information-theoretic?

$$\underbrace{\sqrt{\frac{Rd + R^3}{n}}}_{\text{minimax lower bound}} \quad \text{vs.} \quad \underbrace{\sqrt{\frac{Rd + R^3}{n}} + \overbrace{\frac{R^2d}{n}}^{\text{probably necessary for MLE}}}_{\text{MLE upper bound}}.$$

Remaining question

Without restricting to the MLE, is the extra term $\frac{R^2d}{n}$ information-theoretically necessary?

Two possibilities:

- Necessary $\implies$ prove an R^2d/n lower bound;
- Unnecessary $\implies$ construct an estimator that removes it.

We show that the extra R^2d/n term is **not** information-theoretically necessary and is removed by an *efficient* estimator

Sample-Splitting MLE

Split the data into two blocks $\mathcal{I}_1, \mathcal{I}_2$, with $m_j = |\mathcal{I}_j|$.

Step 1: estimate direction On $\mathcal{I}_1$, compute

$$\check{\theta} \in \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{m_1} \sum_{i \in \mathcal{I}_1} \left[\log(1 + e^{x_i^\top \theta}) - y_i x_i^\top \theta \right].$$

Let $\check{v} := \check{\theta} / \|\check{\theta}\|_2$, and by Chardon et al (2024) $\|\check{v} - v^*\|_2 \lesssim \sqrt{d/(nR)}$ (3)

Step 2: estimate norm On $\mathcal{I}_2$, compute the one-dimensional MLE

$$\hat{r} \in \arg \min_{r > 0} \frac{1}{m_2} \sum_{i \in \mathcal{I}_2} \left[\log(1 + e^{r x_i^\top \check{v}}) - y_i r x_i^\top \check{v} \right].$$

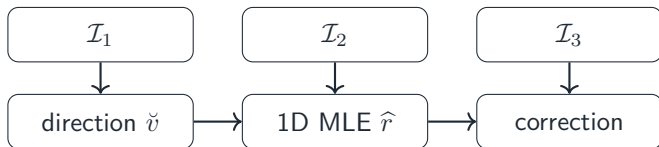
Population bias $R^2 d/n$

Its population minimizer is $r^\circ = q^{-1}(\langle \check{v}, v^* \rangle q(R))$. Therefore

$$\begin{aligned} |\hat{r} - R| &\leq \underbrace{|r^\circ - R|}_{\substack{\asymp R^3 \|\check{v} - v^*\|_2^2 \\ \lesssim R^2 d/n \\ \text{population bias}}} + \underbrace{|\hat{r} - r^\circ|}_{\tilde{O}(\sqrt{R^3/n})}. \end{aligned} \quad (4)$$

$$(3) + (4): \|\hat{r}\check{v} - \theta^*\|_2 = \tilde{O}\left(\sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3}{n}} + \frac{R^2 d}{n}\right)$$

Debiasing via Sample Splitting



$$\underbrace{q(R)^2}_{q(r) \nearrow} = \underbrace{\langle \check{v}, v^* \rangle^2 q(R)^2}_{:= q(r^\circ)^2 \approx q(\hat{r})^2} + \underbrace{(1 - \langle \check{v}, v^* \rangle^2) q(R)^2}_{\substack{\text{missing!} \\ \text{constructed using } \mathcal{I}_3}}.$$

An Initial Construction

Let $P_{\check{v}}^\perp = I_d - \check{v}\check{v}^\top$. We want

$$\underbrace{(1 - \langle \check{v}, v^* \rangle^2)}_{\|P_{\check{v}}^\perp v^*\|_2^2 = 1 - \langle \check{v}, v^* \rangle^2} \times \underbrace{q(R)^2}_{\mathbb{E}(y_i x_i) = q(R) v^*}.$$

Construct $w_{i0} := P_{\check{v}}^\perp y_i x_i$, $i \in \mathcal{I}_3$, so that

$$\mathbb{E} w_{i0} = P_{\check{v}}^\perp \mathbb{E}(y_i x_i) = q(R) P_{\check{v}}^\perp v^*.$$

Further construct the U-statistic

$$\hat{B}_0 = \frac{1}{|\mathcal{I}_3|(|\mathcal{I}_3| - 1)} \sum_{i \neq j} \langle w_{i0}, w_{j0} \rangle,$$

and notice

$$\begin{aligned} \mathbb{E} \hat{B}_0 &= \langle \mathbb{E} w_{i0}, \mathbb{E} w_{j0} \rangle = \|q(R) P_{\check{v}}^\perp v^*\|_2^2 \\ &= (1 - \langle \check{v}, v^* \rangle^2) q(R)^2. \end{aligned}$$

Does this work?

Variance Reduction

Recall

$$w_{i0} = P_{\check{v}}^\perp y_i x_i, \quad \mathbb{E} w_{i0} = q(R) P_{\check{v}}^\perp v^*.$$

In light of $\mathbb{E} y_i = s(R x_i^\top v^*)$, $\check{v} \approx v^*$, $\hat{r} \approx R$,

Final construction is

$$w_i := P_{\check{v}}^\perp \left[y_i - s(\hat{r} x_i^\top \check{v}) \right] x_i, \quad i \in \mathcal{I}_3,$$

$$\begin{aligned} \mathbb{E} w_i &= P_{\check{v}}^\perp \left[\mathbb{E}(y_i x_i) - \mathbb{E}(s(\hat{r} x_i^\top \check{v}) x_i) \right] \\ &= q(R) P_{\check{v}}^\perp v^*, \end{aligned}$$

$$\hat{B} = \frac{1}{|\mathcal{I}_3|(|\mathcal{I}_3| - 1)} \sum_{i \neq j} \langle w_i, w_j \rangle.$$

and notice

$$\begin{aligned} \mathbb{E} \hat{B} &= \langle \mathbb{E} w_i, \mathbb{E} w_j \rangle = \left\| q(R) P_{\check{v}}^\perp v^* \right\|_2^2 \\ &= (1 - \langle \check{v}, v^* \rangle^2) q(R)^2. \end{aligned}$$

Minimax optimal estimator

- $\mathcal{I}_1$: compute the MLE and keep only its direction $\check{v}$.
- $\mathcal{I}_2$: with $\check{v}$ fixed, compute the misspecified one-dimensional MLE $\hat{r}$.
- $\mathcal{I}_3$: construct the correction $\hat{B}$ for the missing orthogonal signal.

Use $\hat{S} := q(\hat{r})^2 + \hat{B}$ as an estimate for $q(R)^2$, then convert back from $q(R)^2$ to R :

$$\hat{q} := \Pi_{[q(\hat{r}/2), q(2\hat{r})]} \left(\sqrt{[\hat{S}]_+} \right),$$

$$\hat{R}_{\text{DB}} := q^{-1}(\hat{q}).$$

Minimax upper bounds; Theorems 2.3, 3.3 of Chen–Mazumdar (2026b)

If the three blocks have comparable sizes and $n \gtrsim R(d+t)$, then with high probability

$$|\hat{R}_{\text{DB}} - R| \lesssim \sqrt{\frac{R^3 t}{n}}.$$

$$\|\hat{R}_{\text{DB}} \hat{v}_{\text{MLE}} - \theta^*\|_2 \lesssim \sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3 t}{n}}.$$

Proof Overview

- ① **Convex localization.** Show $|\hat{r} - r^\circ| \lesssim \sqrt{R^3/n}$ by bracketing the minimizer:

$$\ell'(r^\circ - C\sqrt{\frac{R^3}{n}}) < 0 < \ell'(r^\circ + C\sqrt{\frac{R^3}{n}}).$$

- ② **Concentration of the correction.** Bernstein's and Hanson–Wright inequalities give

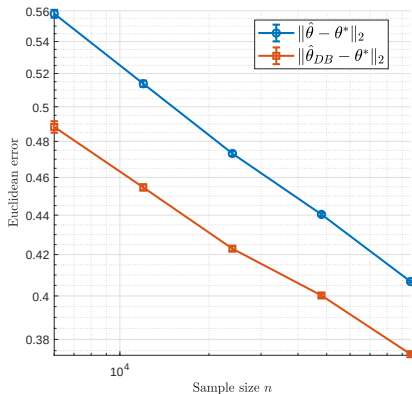
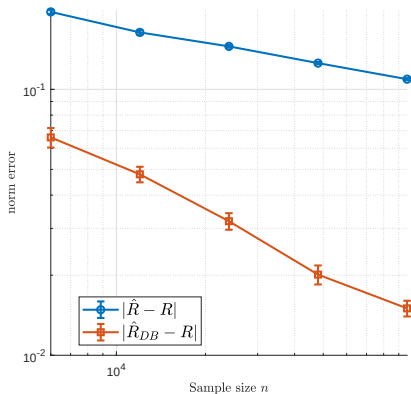
$$|\hat{B} - q(R)^2(1 - \langle \check{v}, v^* \rangle^2)| \lesssim \frac{\sqrt{d}}{nR}.$$

Specifically, for some zero-mean η_i 's, $e_i := y_i - s(\hat{r}x_i^T \check{v})$, and iid $\tilde{g}_i \sim N(0, I_{d-2})$,

$$\begin{aligned} & \hat{B} - q(R)^2(1 - \langle \check{v}, v^* \rangle^2) \\ &= \underbrace{\frac{C_1}{m} \sum_{i=1}^m \eta_i + \frac{1}{m(m-1)} \left[\left(\sum_{i=1}^m \eta_i \right)^2 - \sum_{i=1}^m \eta_i^2 \right]}_{\text{Bernstein's}} \\ &+ \underbrace{\frac{1}{m(m-1)} \sum_{i \neq j} e_i e_j \tilde{g}_i^T \tilde{g}_j}_{\substack{\text{Hanson-Wright} \\ \text{(to the stacked Gaussian vector)}}} \end{aligned}$$

- ③ **Basic algebra.** Combine the two bounds to obtain $|\hat{R}_{\text{DB}} - R| \lesssim \sqrt{R^3/n}$.

Numerical Simulations



$R = n^{0.005}$, $d = 0.3n^{0.8}$, so

$$\underbrace{|\hat{R}_{DB} - R|}_{\text{Debiased estimator}} \lesssim n^{-0.4925}$$

$$\underbrace{|\hat{R}_{MLE} - R|}_{\text{MLE}} = \tilde{O}(n^{-0.19}).$$

Global linear convergence with small stepsize

Theorem 1 of Chen–Mazumdar (2026a)

Let $R = \|\theta^*\| \geq 1$, initialize $\theta_0 = 0$, and take a constant

$$\eta \in (0, \underbrace{8}_{\approx \frac{2}{\beta}}).$$

Under $n \gtrsim R^6 d \log n$, with high probability,

$$\|\theta_t - \theta^*\|_2 \leq \left(1 - \frac{c}{R^3}\right)^t R + \tilde{O}\left(\sqrt{\frac{R^5 d}{n}}\right), \quad \forall t \geq 0.$$

- **Linear convergence:** geometric contraction to a statistical floor, rather than only the classical $L_n(\theta_t) - \min_{\theta} L_n(\theta) = O(1/t)$ objective-gap guarantee.
- **Finite-sample estimation:** an explicit error floor $\tilde{O}\left(\sqrt{R^5 d/n}\right)$ — suboptimal in R but is the first such result.
- **Global convergence:** zero initialization can be replaced by any $\theta_0 \in \mathbb{R}^d$ without essential cost.

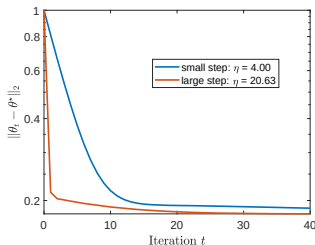
Large-stepsize GD: a local acceleration

Theorem 2 of Chen–Mazumdar (2026a)

Let $R = \|\theta^*\| \geq 1$, initialize θ_0 such that $\|\theta_0 - \theta^*\|_2 \lesssim R^{-1}$. $\eta = \frac{R}{q(R)}$. Under $n \gtrsim R^6 d \log n + R^7 d$, with high probability,

$$\|\theta_t - \theta^*\|_2 \leq \left(1 - \frac{c}{R^2}\right)^t R + \tilde{O}\left(\sqrt{\frac{R^5 d}{n}}\right), \quad \forall t \geq 0.$$

- **Gain:** faster linear convergence — contraction rate $1 - \Theta(R^{-2})$.
- **Cost:** warm initialization.
- **The same statistical error.**



$(n, d, R) = (80000, 100, 8)$, $\|\theta_0 - \theta^*\|_2 = 1$.

Proof overview: Approximate invertibility condition (AIC)

$$\begin{aligned} h_{\theta^*}(u) &= \nabla L_n(u) = \frac{1}{n} \sum_{i=1}^n (s(x_i^\top u) - y_i) x_i \\ &= \underbrace{\frac{1}{n} \sum_{i=1}^n (s(x_i^\top u) - s(x_i^\top \theta^*)) x_i}_{:= h_{\theta^*}^{(1)}(u)} + \underbrace{\frac{1}{n} \sum_{i=1}^n (s(x_i^\top \theta^*) - y_i) x_i}_{:= h_{\theta^*}^{(2)}(u)}. \end{aligned}$$

AIC

$\forall u \in B_2^d(2R).$

$$\|u - \theta^* - \eta h_{\theta^*}(u)\|_2 \leq \left(1 - \frac{c}{R^3}\right) \|u - \theta^*\|_2 + C \left(\sqrt{\frac{d}{nR}} + \frac{d}{n} \right),$$

$$\begin{aligned} &\|u - \theta^* - \eta h_{\theta^*}(u)\|_2 \\ &\leq \eta \|h_{\theta^*}^{(1)}(u) - \mathbb{E}[h_{\theta^*}^{(1)}(u)]\|_2 \quad \text{localization: peeling \& covering} \\ &\quad + \eta \|h_{\theta^*}^{(2)}(u) - \mathbb{E}[h_{\theta^*}^{(2)}(u)]\|_2 \quad \text{Bernstein \& covering} \\ &\quad + \|u - \theta^* - \eta \mathbb{E}[h_{\theta^*}(u)]\|_2 \quad \text{linear algebra.} \end{aligned}$$

Friedlander–Jeong–Plan–Yilmaz (2021); Matsumoto–Mazumdar (2024, 2025);
Chen–Yuan (2024, 2026); Chen–Ding–Xia–Yuan (2025); Chen–Maleki (2026); ...

Summary of Upper Bounds

Estimator & result	Statistical error	Sample complexity	Notes
MLE Chardon et al. (2024)	$O\left(\sqrt{\frac{R^3 d}{n}}\right)$	$\Omega(Rd)$	Best-known result before our work
MLE Chen–Mazumdar (2026b)	$\tilde{O}\left(\sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3}{n}} + \frac{R^2 d}{n}\right)$	$\Omega(Rd)$	Best-known result; conjectured to be tight
MLE direction + debiased norm Chen–Mazumdar (2026b)	$O\left(\sqrt{\frac{Rd}{n}} + \sqrt{\frac{R^3}{n}}\right)$	$\Omega(Rd)$	Minimax optimal
GD Chen–Mazumdar (2026a)	$\tilde{O}\left(\sqrt{\frac{R^5 d}{n}}\right)$	$\tilde{\Omega}(R^6 d)$	Linear convergence

Open Questions

- ① **Gradient descent:** improved sample complexity and error rate.
- ② **MLE:** a finite-sample lower bound $\Omega(R^2 d/n)$.
- ③ **Beyond Gaussian design:** extension to more general covariates.
- ④ **Robustness:** misspecification and corruption.
- ⑤ **High-dimensional structured parameters:**
sparse vectors and low-rank matrices/tensors.

Thank you

Questions?

jc6315@columbia.edu | <https://junrenchen58.github.io/>

Selected references I



J. Chen and A. Mazumdar.

Finite-Sample Performance of Gradient Descent in Logistic Regression with Gaussian Design.

arXiv:2606.21683, 2026.



J. Chen and A. Mazumdar.

Minimax Optimal Estimator and Improved Error Rate for the MLE in Logistic Regression with Gaussian Design.

arXiv:2608.17260, 2026.



H. Chardon, M. Lerasle, and J. Mourtada.

Finite-Sample Performance of the Maximum Likelihood Estimator in Logistic Regression.

arXiv:2411.02137, 2024.



D. Hsu and A. Mazumdar.

On the Sample Complexity of Estimation in Logistic Regression.

COLT, 2024.

Selected references II



P. Sur and E. J. Candès.

A Modern Maximum-Likelihood Theory for High-Dimensional Logistic Regression.
PNAS, 2019.



E. J. Candès and P. Sur.

The Phase Transition for the Existence of the Maximum Likelihood Estimate in High-Dimensional Logistic Regression.
Annals of Statistics, 2020.



D. Ostrovskii and F. Bach.

Finite-Sample Analysis of M-Estimators Using Self-Concordance.
Electronic Journal of Statistics, 2021.



F. Kuchelmeister and S. van de Geer.

Finite Sample Rates for Logistic Regression with Small Noise or Few Samples.
Sankhya A, 2024.